

Name: _____

MATLAB commands you might use
`cos(x)` : cosine of x
`sin(x)` : sine of x
`rand(sz1, sz2)` : returns an `sz1` x `sz2` matrix with random values in the segment (0.0, 1.0).
- if you want values in the range (a,b), where $a < b$, `rand(sz1, sz2) * (b - a) + a`.

1 Array Indexing (2 pts)

Given the following matrix A :

$A =$

75	3	31	79	75
74	27	95	18	27
39	4	3	48	67
65	9	43	44	65
17	82	38	64	16
70	69	76	70	11

What's the result of $A(15)$?

2 Multidimensional Arrays (4 pts)

In computer graphics, a rotation of a point $p = [x, y, z]^T$ around the z -axis by an angle $\theta \in [0, 2\pi)$ is given by the 3×3 rotation matrix:

$$R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Given $\theta = \frac{\pi}{4}$, write a MATLAB expression that stores this 3×3 rotation matrix in the variable R .

3 Array and Matrix Operations (4 pts)

Write MATLAB code to compute $z = Wx$, where W is a 10×784 matrix which has random values in the range $(-0.1, 0.1)$, and x is a 784×1 column-vector which has random values in the range $(0, 255.0)$